

```
def check_subset_sum(numbers, target, subset):
```

```
    total = sum(subset)
```

```
    if total == target:
```

```
        return True
```

```
    else:
```

```
        return False
```

```
numbers = [3, 4, 5, 6]
```

```
target = 9
```

```
subset = [4, 5]
```

```
print("Numbers:", numbers)
```

```
print("Target:", target)
```

```
print("Given subset:", subset)
```

```
if check_subset_sum(numbers, target, subset):
```

```
    print("Solution is valid")
```

```
else:
```

```
    print("Solution is not valid")
```

```
print()
```

```
print("P          : Problems solvable in polynomial time")
```

```
print("NP         : Solutions can be verified in polynomial time")
```

```
print("NP-Hard    : At least as hard as every problem in NP")
```

```
print("NP-Complete : Problems that are both NP and NP-Hard")
```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56)
on win32

Enter "help" below or click "Help" above for more information

```
>>>
```

```
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Documents/BI
```

```
Numbers: [3, 4, 5, 6]
```

```
Target: 9
```

```
Given subset: [4, 5]
```

```
Solution is valid
```

```
P          : Problems solvable in polynomial time
```

```
NP         : Solutions can be verified in polynomial time
```

```
NP-Hard    : At least as hard as every problem in NP
```

```
NP-Complete : Problems that are both NP and NP-Hard
```

```
>>>
```

```
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```